

Merton (1976): Option Pricing When Underlying Stock Returns Are Discontinuous

Introduction

The Black–Scholes model prices European options assuming stock prices follow a geometric Brownian motion with continuous sample paths — no sudden jumps. While Merton (1973) showed the formula is robust to several other assumptions, continuity of the price path is essential: it is what makes the perfect hedge possible. In practice, however, stock prices respond discontinuously to earnings surprises, regulatory rulings, and macroeconomic shocks, producing return distributions with far heavier tails than the log-normal model predicts. Merton (1976) extends the framework by incorporating a jump component directly into the price process.

The Jump-Diffusion Model

The stock price dynamics are governed by:

$$\frac{dS_t}{S_t} = (\alpha - \lambda k) dt + \sigma dZ_t + dq_t \quad (1)$$

where α is the expected return, σ^2 the diffusion variance (conditional on no jump), dZ_t a standard Wiener process, and dq_t a Poisson jump component with intensity λ (mean jumps per unit time). On each jump the stock moves from S_t to $S_t Y$, where $Y > 0$ is a random multiplier with $k \equiv \mathbb{E}(Y - 1)$; the λk correction keeps the total expected return equal to α . Setting $\lambda = 0$ recovers Black–Scholes exactly. The Poisson process is the natural choice for jump timing: its memoryless property reflects the informational efficiency requirement that the arrival of market-moving news be unpredictable.

Option Pricing and the Breakdown of the Black–Scholes Hedge

When a jump occurs, the stock price moves discontinuously and the option value responds nonlinearly. Because portfolio replication is a linear operation, no combination of stock and bond can hedge this nonlinear jump exposure, making the market *incomplete*. Merton resolves the indeterminacy by assuming jump risk is firm-specific and therefore non-systematic (diversifiable). Under the CAPM, a zero-beta portfolio must earn the risk-free rate, which yields the integro-differential pricing equation:

$$0 = \frac{1}{2}\sigma^2 S^2 F_{SS} + (r - \lambda k) S F_S - F_\tau - rF + \lambda \mathbb{E}\{F(SY, \tau) - F(S, \tau)\} \quad (2)$$

with $F(0, \tau) = 0$ and $F(S, 0) = \max[0, S - E]$. The equation is preference-free and

reduces to the Black–Scholes PDE when $\lambda = 0$. For log-normally distributed jump sizes with $\mathbb{E}(Y) = 1$ and $\text{Var}(\ln Y) = \delta^2$, the closed-form solution is:

$$F(S, \tau) = \sum_{n=0}^{\infty} \frac{e^{-\lambda\tau} (\lambda\tau)^n}{n!} W\left(S, \tau; E, \sigma^2 + \frac{n\delta^2}{\tau}, r\right) \quad (3)$$

where $W(\cdot)$ is the Black–Scholes formula. The price is a Poisson-weighted average of Black–Scholes prices, one for each possible number of jumps n , each evaluated at an inflated variance $\sigma^2 + n\delta^2/\tau$ that reflects the additional uncertainty from n jumps.

Implications, Strengths, and Limitations

Equation (3) explains the *volatility smile*: deep out-of-the-money options are underpriced by Black–Scholes because a single large jump can push the stock past the strike instantly, a scenario a continuous diffusion nearly cannot produce over short horizons. This generates the well-known pattern of implied volatilities that rise away from the money. More broadly, the jump-diffusion process produces excess kurtosis consistent with empirical return distributions.

The model’s key strengths are its analytical tractability (the series converges rapidly and each term is a Black–Scholes price), its clear economic interpretation (Poisson arrivals reflect unpredictable firm-specific news), and its role as the foundation for a large family of later models. Its main limitations are the constant jump intensity λ (unable to capture time-varying crash risk or volatility clustering), the symmetric log-normal jump distribution (unable to match the steep downside skew of index options), and the absence of stochastic volatility, all of which are addressed in extensions such as Bates (1996).

Relevance to QQQ Options

The QQQ ETF tracks the NASDAQ-100, a technology-heavy index particularly susceptible to jump-like behaviour. The implied volatility surface extracted from QQQ options displays a pronounced downside skew — out-of-the-money puts carry significantly higher implied volatilities than at-the-money options — consistent with equation (3). We calibrate $(\sigma, \lambda, \delta^2)$ to the observed QQQ cross-section and compare the fit to Black–Scholes. The jump-diffusion model is expected to outperform for short-maturity, far-from-the-money options where jump risk is most material, and to converge to the Black–Scholes fit for longer-dated near-the-money contracts.